Pinot: Change add column semantic

link to doc

Current situation

- Pinot supports schema evolution by adding new columns
- When a new column is added, Servers prune old segments in an inconsistent way

Example

Segment 1

col1	col2
v111	v121
v112	v122

Segment 2

col1	col2	col3
v211	v221	true
v212	v222	false

query	result
select col1 from t1	(v111), (v112), (v211), (v212)
select count(id) from t1	4

👍

query	result
select col3 from t1	(true), (false)
select count(col3) from t1	4

🤨

query	result
select * from t1	(v211, v221, true), (v212, v222, false)
select count(*) from t1	4

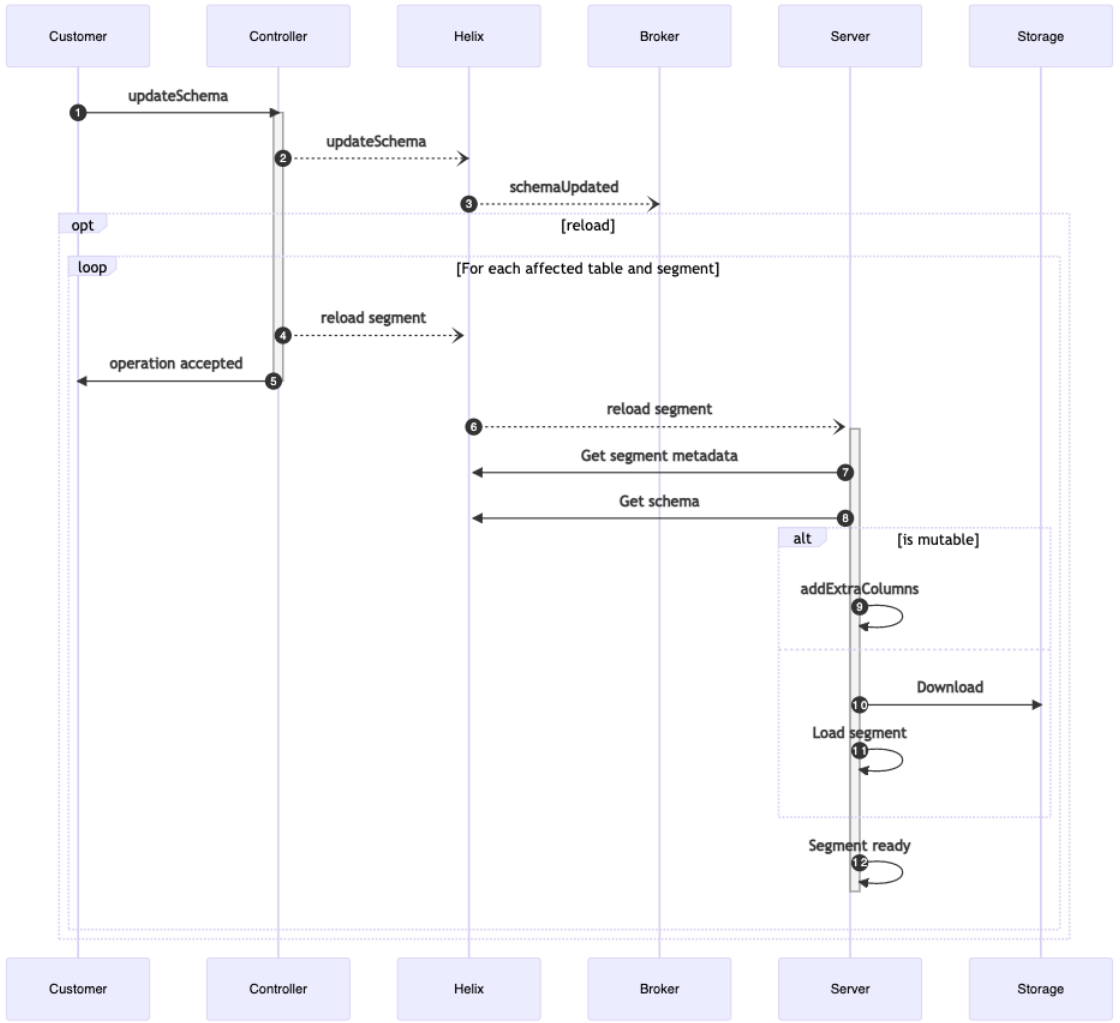
🤔

query	result
select count(*) from t1	4
select count(*) from t1 where col3 is not null	2
select count(*) from t1 where col3 is not null and col3 is null	2

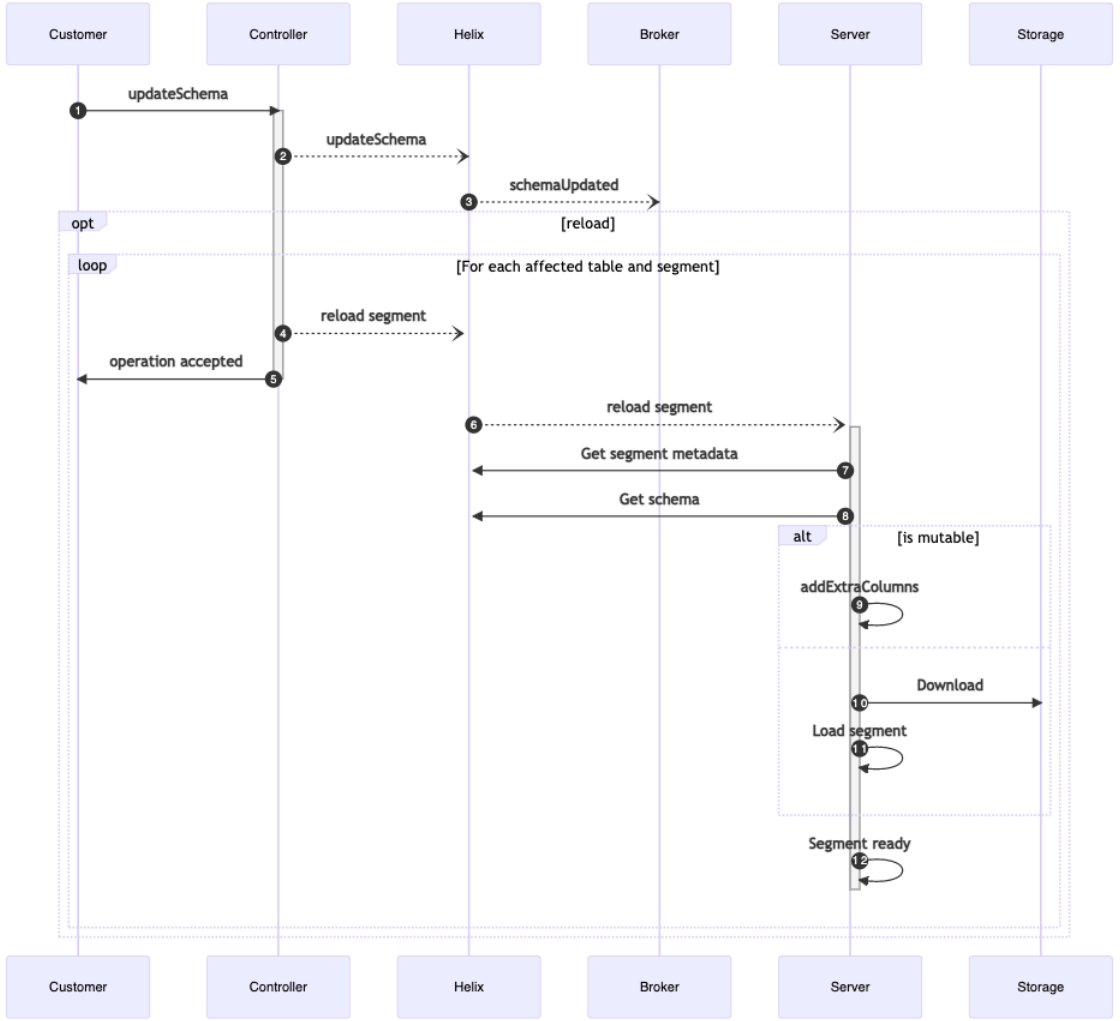


This behavior is unexpected, which could be problematic for users.

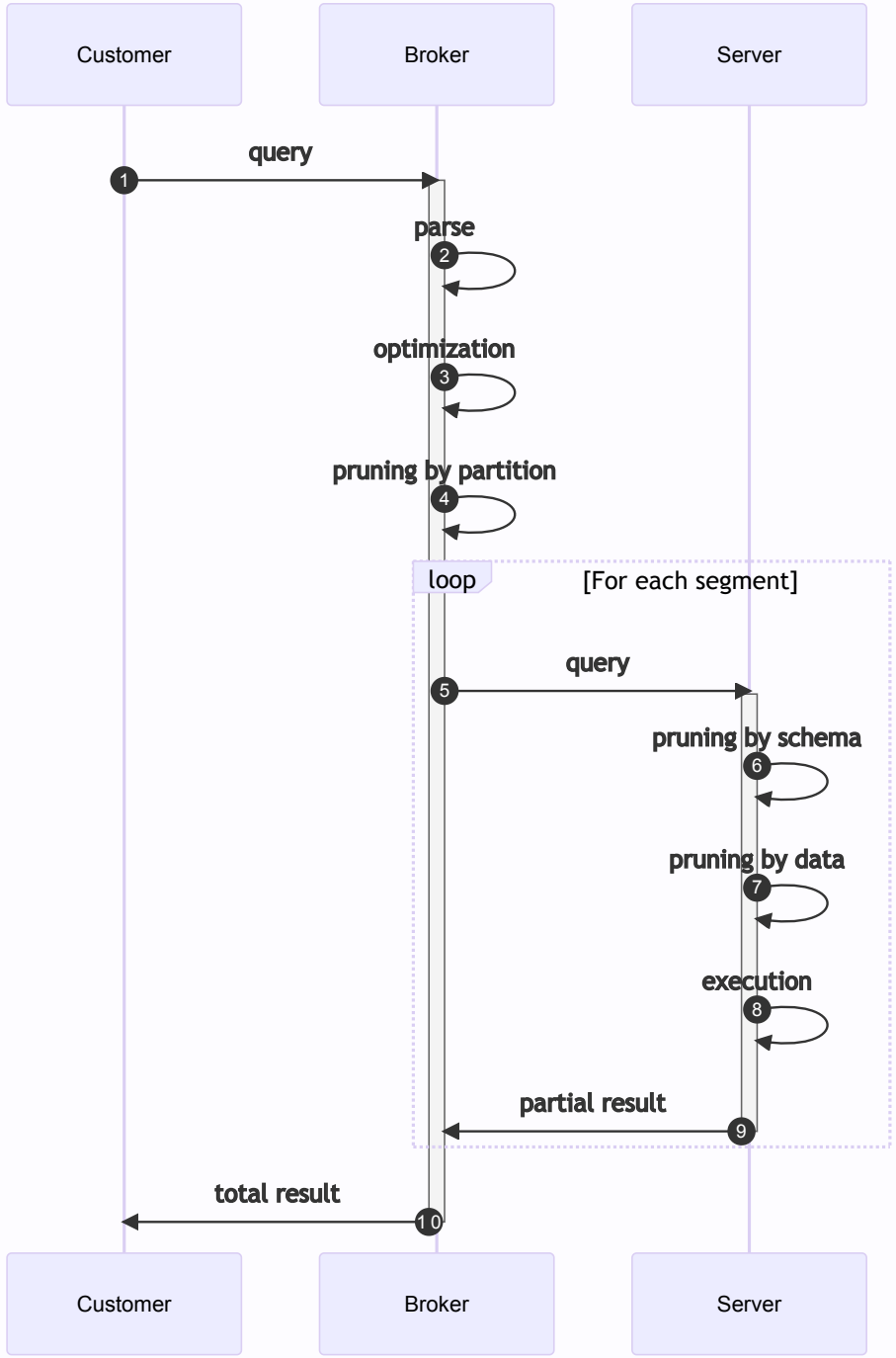
Current implementation



What happens if reload is not specified?



What happens between 3 and 12?


Customer
Broker
Server
query
1
parse
2
optimization
3
pruning by partition
4
loop
[For each segment]
query
5
pruning by schema
6
pruning by data
7
execution
8
partial result
9
total result
10
Customer
Broker
Server

Effective schema

vs

Physical schema

New semantic

Segment 1

col1	col2
v111	v121
v112	v122

Segment 2

col1	col2	col3
v211	v221	true
v212	v222	false

query	old	new
select col1 from t1	(v111), (v112), (v211), (v212)	(v111), (v112), (v211), (v212)
select count(id) from t1	4	4

👍

query	old	new
select col3 from t1	(true), (false)	(null), (null), (true), (false)
select count(col3) from t1	4	4

👍

query	old	new
select * from t1	(v211, v221, true), (v212, v222, false)	(v111, v112, null), (v121, v122, null), (v211, v221, true), (v212, v222, false)
select count(*) from t1	4	4



query	old	new
select count(*) from t1	4	4
select count(*) from t1 where col3 is not null	2	2
select count(*) from t1 where col3 is not null and col3 is null	2	4

👍

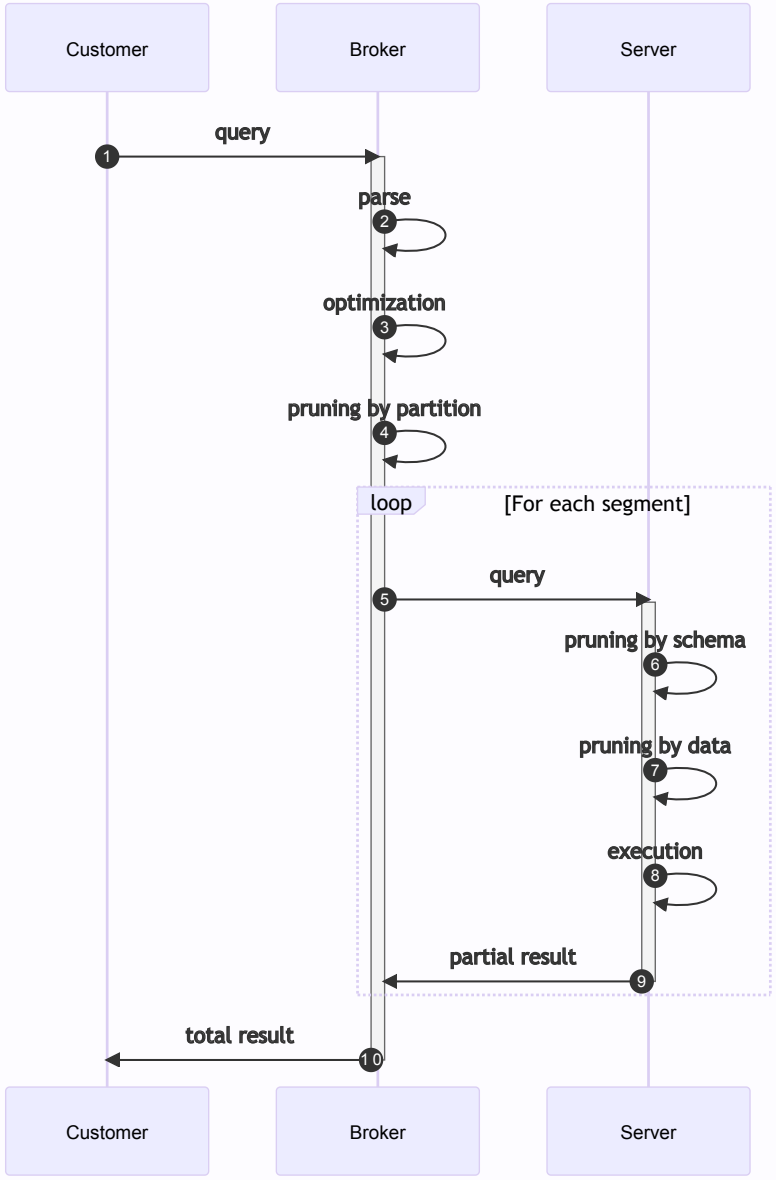
Proposed new implementation

- Servers will store the effective schema for each table
- Servers will validate queries with that schema
  - If a query uses a column that is not in the schema, an error is thrown

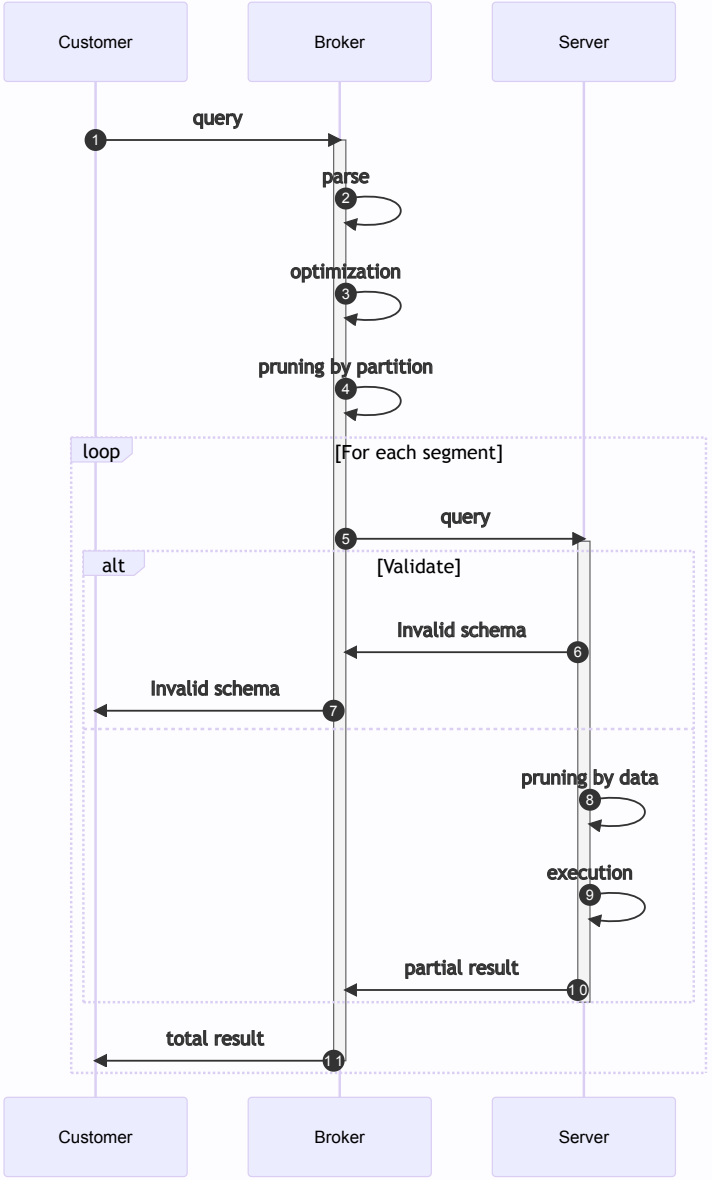
Proposed new implementation

- When segments are loaded, Servers checks whether all columns in the effective schema are in the physycal schema or not.
  - If not, a virtual column is created
    - For each cell, this column returns the default null value on the effective schema

old

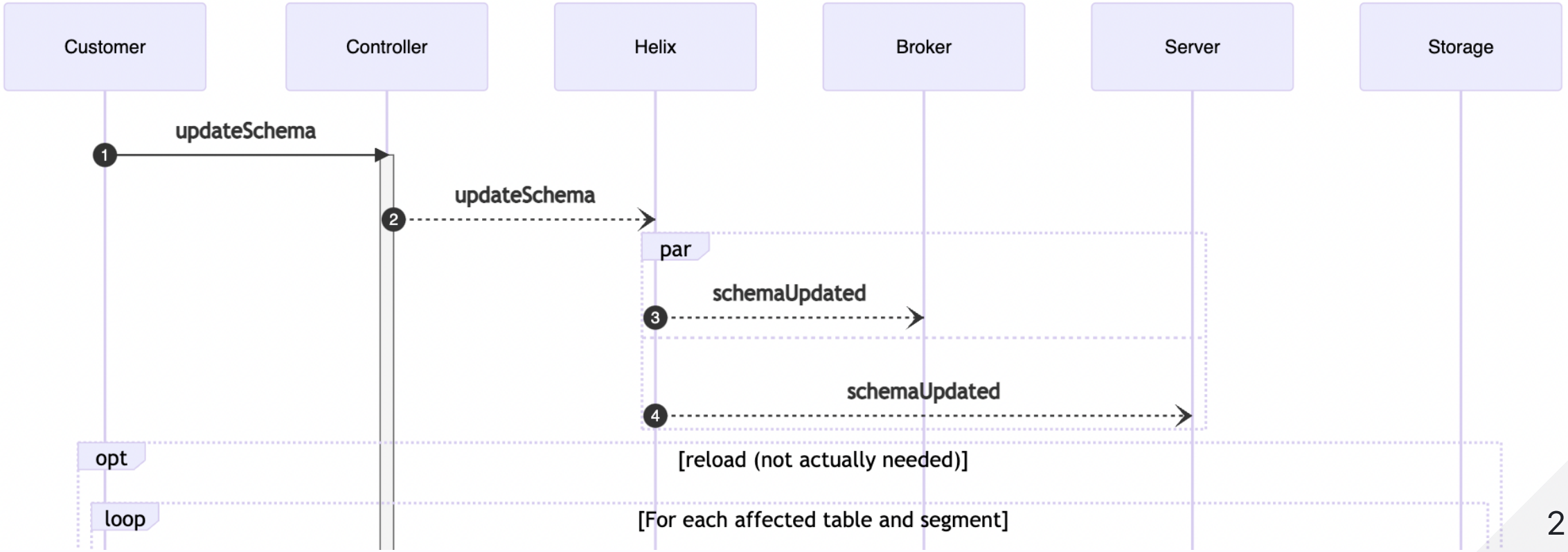

Customer
Broker
Server
query
parse
optimization
pruning by partition
loop
[For each segment]
query
pruning by schema
pruning by data
execution
partial result
total result
Customer
Broker
Server

new


Customer
Broker
Server
query
parse
optimization
pruning by partition
loop
[For each segment]
query
alt
[Validate]
Invalid schema
Invalid schema
pruning by data
execution
partial result
total result
Customer
Broker
Server

How to do it

- Servers should listen for updateSchema Helix messages



The document also discuss about how this new semantic affects other DDL operations we don't support right now, like remove columns or change their types.

link to doc

Q & A